DEPARTMENT OF AEROSPACE ENGINEERING, SCHOOL OF MECHANICAL & MANUFACTURING ENGINEERING (SMME), NUST, ISLAMABAD

Lab Project

The Mini FADEC System

Electronics Engineering (EE-227)

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CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABSTRACT

This project presents the design and development of a Mini FADEC (Full Authority Digital Engine Control) system that simulates aircraft engine control using a DC motor. The system processes throttle input through a potentiometer and uses PWM to regulate motor speed. A temperature sensor is integrated to monitor system conditions and enforce safety mechanisms such as speed reduction and shutdown under overheating. The project highlights real world engineering challenges including hardware-software integration, grounding issues, and power management. Despite implementation challenges, the system demonstrates core FADEC principles of intelligent control, safety, and efficiency.

1- INTRODUCTION TO THE LAB PROJECT

1.1- Project Overview & Learning Objectives

This lab project focuses on the design and implementation of an embedded control system using discrete electronic components and an Arduino microcontroller. The system integrates sensor input, signal processing, and actuator control to simulate a real world engineering application.

In this project, a microcontroller based system is developed to read input signals (such as throttle position and temperature), process the data, and control a motor accordingly. The implementation combines both hardware and software aspects, allowing students to gain hands on experience in circuit design, sensor interfacing, and embedded programming.

Figure No. 01: The FADEC Setup

The key learning objectives of this project include:

- Developing an understanding of embedded systems and microcontroller based design.
- Applying fundamental electronics concepts such as resistors, diodes, and motor drivers.
- Interfacing sensors and actuators with a microcontroller.
- Implementing control logic using programming.
- Gaining practical troubleshooting and system integration experience.

This project bridges theoretical knowledge with practical implementation, preparing students for real world engineering challenges.

1.2- Problem Statement & Problem Significance

The objective of this project is to design a Mini FADEC based smart control system that can regulate motor speed based on user input while ensuring safe operation through temperature monitoring.

In real world aircraft systems, Full Authority Digital Engine Control (FADEC) is used to manage engine performance efficiently and safely. Inspired by this concept, this project aims to simulate a simplified FADEC system using an embedded platform.

The problem addressed in this project is:

“How to design an intelligent control system that can respond to user input while preventing system damage due to unsafe operating conditions such as overheating.”

This project is significant because it demonstrates:

- The importance of automation and intelligent decision making systems.
- The role of safety mechanisms in engineering design.
- The integration of electronics, programming, and control systems.

It also highlights how real world aerospace concepts can be simplified and implemented at a laboratory scale.

1.3- Expected Outcomes & Deliverables

The successful completion of this project requires both a functional system and comprehensive technical documentation. The expected outcomes include the development of a working prototype along with proper engineering analysis and presentation.

The deliverables for this project are:

- A functional embedded system demonstrating motor control with safety features.
- A detailed and well structured project report following standard formatting guidelines.
- Well commented Arduino source code explaining system logic.
- Professional circuit schematics designed using appropriate software tools.
- A formal presentation explaining the design, working, and results of the project.
- A project development logbook included as an annexure in the report.

2- SCOPE OF THE REPORT

This report covers the design, development, and analysis of a Mini FADEC based embedded control system. It includes system architecture, hardware components, circuit design, and software implementation using a microcontroller. The report also discusses working principles, testing procedures, challenges encountered, and system limitations. The scope is focused on demonstrating the integration of electronics and control logic to simulate the functionality of a Full Authority Digital Engine Control system at a laboratory scale.

3- GENERAL PROJECT PLANNING

3.1- Group Formation & Roles

The project was carried out in a group of four members, with each member actively contributing to all phases of the project. To ensure balanced skill development and effective collaboration, responsibilities were distributed among team members in roles such as:

- **Lead Engineer** (overall system design and integration): Wajeeha Zubair
- **Software Lead** (Arduino programming and logic development): Aleena Kabir
- **Hardware Lead** (circuit design and component interfacing): Ayesha Yaseen
- **Documentation Lead** (report writing and record keeping): Ayesha Choudhary

Role rotation was encouraged throughout the project to ensure that all members gained exposure to both hardware and software aspects. Team performance was evaluated based on coordination, contribution, and effective workload distribution.

3.2- Project Workflow & Development Phases

A structured, phase wise approach was followed to manage the project efficiently:

- **Phase 1: Project Proposal**

In this phase, the problem statement was defined, and the system concept (Mini FADEC based motor control) was finalized. Preliminary circuit designs were developed, required components were identified, and the overall software logic was outlined. The proposal was formally approved by the course instructor before proceeding further.

- **Phase 2: Components Acquisition & Initial Prototyping**

All required components, including sensors, motor driver, and microcontroller, were acquired. Individual modules such as the DC motor, potentiometer, and temperature sensor were tested separately using breadboard setups to verify their functionality.

- **Phase 3: System Integration & Development**

In this phase, the complete circuit was assembled, and Arduino code was developed to implement system logic. Integration of input (throttle), processing (Arduino), and output (motor control) was performed. Debugging and iterative improvements were carried out to achieve desired functionality.

- **Phase 4: Testing, Refinement, & Documentation**

The system was tested under different conditions to evaluate performance. Issues related to power distribution, grounding, and signal behavior were identified and resolved. Final refinements were made, and complete documentation, including the report and presentation, was prepared.

3.3- Documentation Requirements

All submitted materials were prepared following professional engineering standards:

- **Project Report**

The report is structured according to specified guidelines, using *Times New Roman* font (size 12), 1.5 line spacing, and proper alignment. It clearly explains the design, implementation, and analysis of the system.

- **Arduino Source Code**

The code is well commented, with clear explanations of logic and control decisions. Proper formatting, indentation, and meaningful variable names are used to ensure readability.

- **Circuit Schematics**

Professional circuit diagrams are included, clearly showing all components, interconnections, and signal flow.

- **Tables & Figures**

All data is presented using properly labeled tables and figures. Each visual element is numbered, captioned, and referenced within the text for clarity.

3.4- Presentation Preparation

A formal presentation was prepared to demonstrate the project. It includes:

- Explanation of system design and working principle
- Justification of design choices
- Demonstration of system functionality
- Discussion of challenges and results

The focus was on clear communication, technical understanding, and confident delivery.

3.5- Submission Schedule & Milestones

The project followed a structured timeline with defined milestones to ensure timely completion. Key deliverables included:

- Project Proposal (initial design and planning)
- Interim progress updates (if applicable)
- Final Project Report (complete documentation)
- Arduino Source Code (well structured and commented)
- Functional Prototype Demonstration
- Final Presentation with Q&A

Each milestone contributed to the overall project evaluation, with major weight assigned to system functionality and final report quality.

4- WORKING PRINCIPLE

The Mini FADEC based control system operates by continuously reading input signals, processing them, and controlling the motor accordingly. The throttle input, provided through a potentiometer, is read by the Arduino as an analog signal and mapped to a PWM value to control motor speed. At the same time, the temperature sensor monitors system conditions and sends real time data to the microcontroller. Based on predefined temperature thresholds, the system applies decision making logic: under normal conditions, the motor follows the throttle input; at elevated temperatures, the speed is reduced; and at critical levels, the system shuts down to prevent damage. This continuous loop of input, processing, and output simulates the behavior of a Full Authority Digital Engine Control system, ensuring safe and efficient operation.

5- REQUIRED APPARATUS

The apparatus is given below:

- Arduino Uno
- Jumper wires (male-to-male / male-to-female)

- Potentiometer (for throttle input)
- LM35 temperature sensor
- DC motor
- Motor driver module (BTS 7960)
- Flyback diode
- Buck converter (for voltage regulation)
- Power supply (Cells)
- Fan (to attach with motor)
- 3D printed engine casing
- Resistors (for LEDs and signal conditioning)
- LEDs (for indication, if used)
- USB cable (for Arduino programming)











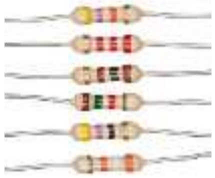

Figure No. 02: The Required Apparatus

6- SYSTEM DESIGN

The details are discussed below in this regard:

6.1- Overall System Architecture

The Mini FADEC system is designed using a modular architecture consisting of input, control, monitoring, output, and safety subsystems. The Arduino Uno acts as the central control unit, processing input signals from the potentiometer and temperature sensor, and generating PWM signals to control motor speed via the motor driver. This structured approach ensures efficient system integration and reliable operation.

6.2- Input System

The potentiometer is used to simulate throttle input. It operates as a voltage divider, producing a variable analog voltage (0-5V) depending on the knob position. This voltage is read by the Arduino through an analog input pin and represents the desired engine power level.

6.3- Control unit (Arduino-FADEC Logic)

The Arduino Uno serves as the central processing unit of the system. It continuously reads input signals, processes control logic, and generates PWM outputs to regulate motor speed. The embedded program simulates FADEC behavior by adjusting output based on throttle input and temperature conditions.

6.4- Monitering System

The LM35 temperature sensor is used to monitor system temperature in real time. It provides an analog voltage proportional to temperature (10 mV/°C), which is read by the Arduino. This data is used to implement safety control logic.

6.5- Output System

The LM35 temperature sensor is used to monitor system temperature in real time. It provides an analog voltage proportional to temperature (10 mV/°C), which is read by the Arduino. This data is used to implement safety control logic.

6.6- Safety System

The safety system is implemented through software based logic. When the temperature exceeds predefined thresholds:

- Motor speed is reduced at moderate temperatures
- Motor is completely shut down at critical temperatures

This ensures safe operation and prevents system damage, similar to real FADEC systems.

7- HARDWARE COMPONENTS

The details are mentioned below:

Table N0. 01: The Hardware Components

Sr. No.	Component	Function
01.	Arduino Uno	Central control unit executing FADEC logic
02.	Potentiometer	Provides throttle input
03.	LM35 Temperature Sensor	Measures temperature
04.	BTS 7960 Motor Driver	Controls motor using PWM
05.	DC Motor	Simulates engine
06.	Diode	Protects against back EMF
07.	Buck Converter	Provides regulated power supply

8- SOFTWARE DESIGN

The details are given below:

8.1- Control Logic

The system continuously reads analog inputs from the potentiometer (0-1023) and maps them to PWM output values (0-255) to control motor speed.

8.2- Decision Making

The Arduino applies conditional logic based on temperature:

- Normal condition: Motor follows throttle input
- Elevated temperature: Motor speed is reduced
- Critical temperature: Motor shuts down

Control decisions are implemented using embedded C++ logic within the Arduino environment.

9- 3D PRINTED ENGINE ASSEMBLY

The step wise explanation is mentioned below:

9.1- Engine Casing Design

To enhance realism and structural design, a 3D printed engine casing was developed to house the DC motor and fan assembly. This allowed the system to visually and functionally resemble an actual engine.

Figure No. 03: The 3D-Printed Engine Casing

9.2- Mechanical Integration

The motor was securely mounted inside the casing, with a fan attached to its shaft to simulate airflow and thrust. The design ensured proper alignment, reduced vibration, and stable operation.

9.3- Engineering Significance

This feature demonstrates:

- Integration of mechanical and electronic systems.
- Practical prototyping and design skills.
- Real world system modeling approach.

10- DETAILED PROJECT INFORMATION & TECHNICAL IMPLEMENTATION

10.1- Arduino Platform Overview & Setup

The Arduino Uno was used as the central control unit of the system, acting as a Mini FADEC (Full Authority Digital Engine Control). It processes input signals, executes control logic, and generates output signals to control the motor.

The system was programmed using the Arduino IDE, where embedded C++ code was written, compiled, and uploaded to the microcontroller. The program structure is based on two fundamental functions:

- setup(): Initializes input/output pins and system parameters
- loop(): Continuously reads inputs, processes logic, and updates outputs

This structure enables real time monitoring and control of the system.

10.2- Sensor Interfacing & Selection

A temperature sensor, LM35, was selected to monitor system conditions and simulate engine temperature.

The LM35 provides an analog voltage output proportional to temperature, which was connected to one of the Arduino's analog input pins. The sensor was powered using a 5V supply and common ground.

Additionally, a potentiometer was used to simulate throttle input. It was connected as a voltage divider, providing a variable analog signal to the Arduino.

These inputs allowed the system to:

- Read user defined throttle levels
- Monitor temperature in real time

10.3- Motor Control Mechanism

A DC motor was used to simulate engine thrust. Since the Arduino cannot supply sufficient current directly, a motor driver, BTS7960, was used to interface the motor with the microcontroller.

Motor speed was controlled using Pulse Width Modulation (PWM). The Arduino generates a PWM signal, which adjusts the effective voltage supplied to the motor, thereby controlling its speed.

The core concept used here is given as:

$$V\ (avg) = D \times V(max)$$

Where:

- D = Duty cycle
- V (avg) = Effective voltage applied to motor

The system maps throttle input values (0-1023) to PWM values (0-255) to control motor speed.

10.4- Implementation of Core Electronics Concepts

The project demonstrates practical application of fundamental electronics components:

- **Diodes:** A flyback diode was connected across the motor terminals to protect the circuit from back EMF generated during switching.
- **Resistors:** Used for signal conditioning and stable operation of components such as sensors and LEDs.
- **Logic Gates:** Initially, an AND gate was introduced for hardware based safety logic. However, due to incompatibility with PWM signals, this approach was replaced with software-based decision making.
- **Motor Driver and Power Electronics:** The motor driver module ensured safe handling of high current, while a buck converter was used for regulated power supply.
- **Circuit Design Principles:** Special attention was given to grounding, voltage levels, and current requirements to ensure stable system operation.

10.5- Circuit Design & Schematic Representation

The circuit was designed following standard engineering practices. A modular approach was adopted, separating input, control, and output sections.

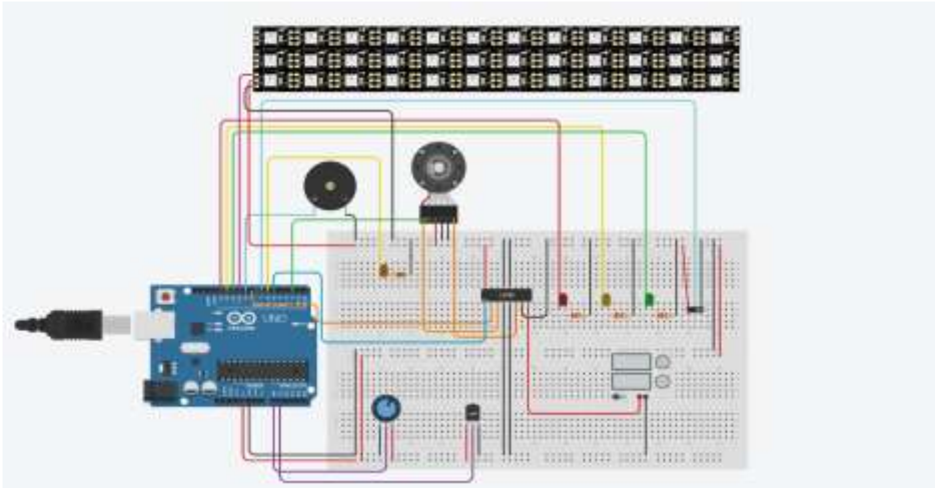

Figure No. 04: The Circuit Diagram for Mini FADEC Project

Key considerations included:

- Proper power distribution
- Common grounding across all components
- Clear separation of signal and power lines

Circuit schematics were prepared using CAD tools to clearly represent all components, interconnections, and signal flow.

10.6- Software Development & Coding Practices

The control logic was implemented using embedded C++ in the Arduino environment.

Key features of the software include:

- Reading analog inputs (throttle and temperature)
- Mapping input values to PWM output
- Conditional decision-making based on temperature thresholds
- Real time control of motor speed

The code was structured with:

- Proper indentation
- Meaningful variable names
- Detailed comments explaining each functional block

The Serial Monitor was used extensively for debugging and monitoring system behavior during development.

10.7- Testing & Debugging Procedures

A systematic approach was followed for testing and debugging:

- **Unit Testing:** Individual components such as the motor, sensor, and potentiometer were tested separately.
- **Integration Testing:** Interfacing between modules (sensor + Arduino + motor driver) was verified.
- **System Testing:** The complete system was tested under different operating conditions, including varying throttle inputs and temperature levels.

The working of the model is mentioned below:

Figure No. 05: The Working Setup

During debugging, several issues were identified and resolved, including:

- Faulty hardware components
- Missing common ground connections
- Power supply instability
- Signal mismatches in hardware logic

Diagnostic tools such as the Arduino Serial Monitor and multimeter were used to isolate and resolve these issues.

11- CHALLENGES & TROUBLESHOOTING

During the development of the Mini FADEC system, several challenges were encountered related to hardware integration and system stability. Initially, the motor failed to operate due to the absence of a common ground between the Arduino and motor driver, which was later corrected. An attempt to implement hardware logic using an AND gate was unsuccessful because PWM signals are time varying, leading to a shift toward software based control. A faulty motor was also identified and replaced. Additionally, system instability occurred when an LCD module was added, mainly due to power distribution issues. A critical short circuit during battery integration caused overheating, highlighting the importance of careful wiring and verification. These issues were resolved through systematic testing, debugging, and redesign, providing valuable insight into real world engineering challenges.

12- LIMITATIONS

The Mini FADEC based system faced several limitations during its implementation, mainly related to hardware stability and system integration. Issues such as power distribution, grounding errors, and component reliability affected overall performance. Additionally, the absence of a fully developed feedback system limited precise control, and time constraints prevented complete optimization of the design.

Key limitations include:

- Hardware instability due to improper power distribution
- Grounding issues affecting signal consistency
- Lack of closed loop feedback for accurate control
- Faulty or unreliable components
- Breadboard connections causing noise and loose contacts
- Limited time for full system testing and optimization

13- FUTURE IMPROVEMENTS

The Mini FADEC based system can be further enhanced by improving hardware reliability, expanding control capabilities, and implementing more advanced feedback mechanisms. With better design optimization and additional sensors, the system can more closely replicate real-world control systems and achieve higher accuracy, stability, and performance.

Future improvements include:

- Implementation of a full closed-loop control system (e.g., PID control)
- Integration of additional sensors such as RPM, pressure, or airflow
- Use of a custom PCB instead of a breadboard for improved reliability
- Enhanced power management using stable regulated power supplies
- Improved motor driver configuration for better efficiency
- Addition of display modules for real time monitoring (LCD/OLED)
- Optimization of code for faster and more stable response
- Development of a more realistic engine model for demonstration

14- RESULTS & DISCUSSION

Individual learning outcomes are discussed below:

- **Ayesha Choudhary:** The developed Mini FADEC system successfully demonstrated the core functionality of controlling motor speed based on throttle input while incorporating temperature-based safety mechanisms. The system responded effectively to changes in the potentiometer, with smooth variation in motor speed achieved through PWM control. Under normal temperature conditions, the motor operated proportionally to the throttle input, validating the mapping logic implemented in the Arduino code. However, when the temperature exceeded predefined thresholds, the system appropriately reduced speed and eventually shut down, confirming the effectiveness of the safety algorithm.

- **Ayesha Yaseen:** From a hardware perspective, the project highlighted both successful integration and practical challenges. The interfacing of components such as the LM35 sensor, potentiometer, and motor driver was achieved, and the system functioned as expected after resolving grounding and wiring issues. Initially, unstable behavior was observed due to improper power distribution and loose breadboard connections, which affected signal consistency. After implementing a common ground and improving wiring, system stability significantly improved.
- **Aleena Kabir:** The software implementation played a crucial role in the success of the Mini FADEC system. The Arduino code effectively processed analog inputs and converted them into meaningful control actions using PWM. The use of conditional statements for temperature based decision making allowed the system to operate safely under different conditions. Replacing hardware logic gates with software-based control proved to be a more flexible and reliable approach, especially for handling PWM signals.
- **Wajeeha Zubair:** This project provided valuable insights into the integration of electronics, programming, and control systems. While the system did not achieve perfect hardware stability, it successfully demonstrated the fundamental working principles of a FADEC system. The challenges faced, including component failures, short circuits, and integration issues, contributed significantly to the learning process. These difficulties highlighted the importance of careful planning, testing, and troubleshooting in engineering design.

15- OVERALL CONCLUSION

This project successfully demonstrated the design and conceptual implementation of a Mini FADEC based control system using an embedded platform. The system was able to process throttle input, monitor temperature, and apply intelligent control to regulate motor speed using PWM. Although full hardware stability and complete integration were not achieved, the project effectively replicated the core working principles of a Full Authority Digital Engine Control system. Throughout the development process, valuable insights were gained regarding sensor interfacing, motor control, power management, and troubleshooting. The challenges faced during implementation highlighted the importance of proper grounding, reliable components, and software-based control logic. Overall, the project provided a strong foundation in embedded systems and demonstrated the practical application of engineering concepts in solving real world problems.

16- REFERENCES

The references are mentioned here:

- Arduino. (2023). *Arduino Uno Rev3*.
- National Instruments. (2020). *Introduction to PWM (Pulse Width Modulation)*.
- Texas Instruments. (2016). *LM35 Precision Centigrade Temperature Sensors Datasheet*.
- NXP Semiconductors. (2018). *Motor Control Fundamentals*.
- Federal Aviation Administration. (2016). *Aircraft Systems: Engine Control Systems (FADEC)*.

17- WORK DISTRIBUTION

The work distribution table is given below:

Table No. 02: The Work Distribution

Member	CMS ID	Work Assigned
Ayesha Choudhary	520024	Introduction, Required

		Apparatus, Engine Assembly, Future Improvements,
Ayesha Yaseen	505483	Scope of the Report, System Design, Technical Implementation, Overall Conclusion,
Aleena Kabir	511170	General Project Planning, Hardware Components, Challenges & Troubleshooting
Wajeeha Zubair	505250	Working Principle, Software Design, Limitations, References

The whole project is considered to be the result of combined sum of the efforts made by all the group members.